

Phan Thanh Cong

Business Analyst

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Professional Summary

Business Analyst with **1+ year of professional experience** in a hybrid BA/developer role, combining requirement analysis with hands-on full-stack development. Direct experience communicating with international clients in **English** to elicit, clarify, and validate requirements across web applications, IoT edge systems, and third-party integrations (Stripe, GoHighLevel).
Strong analytical skills in translating business needs into actionable deliverables: user flows, acceptance criteria, API specifications, and database schemas — with technical depth to evaluate feasibility, identify edge cases, and bridge communication between clients and developers.
JLPT N3 certified. Leverages **AI tools** for accelerated documentation, research, and prototyping.

Core Competencies

Analysis & Documentation: Requirement Elicitation, Functional Requirements, Business Rules, Acceptance Criteria, Use Cases, User Stories, Functional Specifications, Gap Analysis, Impact Analysis, Technical Feasibility Analysis.
Stakeholder & Delivery: Stakeholder Management, Client Communication, Cross-functional Collaboration, Progress Reporting, Agile/Scrum, Backlog Clarification, UAT Support, Defect Clarification, Change Analysis, Release Support.
Process & System Design: User Flows, System Flows, Integration Flows, Data Mapping, State Transitions, Error Handling, Edge-case Analysis, API Documentation.
Technical Understanding: RESTful APIs, Webhooks, Database Structures, Authentication, MQTT, System Integrations, Cloud Deployment, Third-party Services.
Tools & Languages: Postman, Swagger/OpenAPI, DBeaver, Git, Jira, Draw.io, AWS, Docker. English (IELTS 6.0), Japanese (JLPT N3).

Work Experience

Business Analyst / Full-stack Developer, PathTech – Hanoi, Vietnam May 2025 – Present
Hybrid role in a small software company

- Served as the primary BA contact for international clients, conducting weekly requirement sessions in English across 3 concurrent production projects (real estate platform, IoT surveillance system, e-learning platform).
- Translated business requirements into comprehensive API specifications, database schemas, and multi-state workflow definitions for 2 production platforms deployed on AWS.
- Identified 10+ critical edge cases per project — including a race condition in the booking flow that would have caused double-charged customers — preventing defects before production release.
- Assessed change impact across 3+ integrated systems (Stripe, GoHighLevel, MQTT brokers, AWS) and proposed phased rollout strategies to minimize production risk.
- Broke features into 20+ development and testing tasks per sprint, tracked progress across 3 projects, and maintained under 2-day turnaround on requirement clarifications.
- Authored and maintained Swagger/OpenAPI documentation covering 60+ endpoints with request/response examples, and validated integration scenarios using Postman.
- Supported feature validation, UAT, defect clarification, and production issue investigation — applied hands-on development knowledge to explain technical constraints and reduce communication gaps.

Selected Requirement Case

Keanu Residence | Booking, Payment, and CRM Flow Production Project
Requirement Analysis, Client Communication, and Feature Validation

- Received and clarified requirements from international stakeholders for a luxury real estate platform covering property browsing, unit availability, reservations, Stripe payments, lead collection, and CRM follow-up workflows.
- Elicited business rules through structured follow-up questions: unit availability logic, reservation creation triggers, payment-status impact on bookings, and GoHighLevel synchronization scope — reducing ambiguity before development started.
- Mapped the end-to-end flow across a 5-state reservation lifecycle (Pending, Confirmed, Expired, Cancelled, Failed) from unit selection through Stripe checkout, payment confirmation, and CRM synchronization.
- Identified 12+ edge cases across payment and CRM sync flows — including concurrent bookings double-reserving units, delayed Stripe webhooks, duplicate webhook events, and failed CRM synchronization — each documented with expected system behavior.
- Translated clarified requirements into user flows, business rules, API behavior for 60+ endpoints, an 11-table database schema, frontend states, acceptance criteria, and testing scenarios.

- Coordinated requirement clarification across frontend, backend, Stripe, GoHighLevel, and AWS deployment environments — ensuring consistent behavior across all integration boundaries.
- Validated API and integration behavior using Postman, supported feature testing and defect clarification, and reviewed the completed flow against original client expectations before release.
- Produced comprehensive requirement documentation including user flow diagrams, state transition matrices, API behavior specifications, and 30+ acceptance test scenarios — serving as the single source of truth for development and QA.

Technologies involved: ReactJS, NestJS, PostgreSQL, Stripe, GoHighLevel, AWS Amplify, EC2

Other Projects

AI-Powered Smart Surveillance Edge System (PA-CAM) Production Project

Requirement Analysis, System Flow Design, and Integration Delivery

- Analyzed requirements for a 5-state device onboarding lifecycle, real-time telemetry reporting (6 hardware metrics), bidirectional MQTT command protocol, event synchronization, and fault recovery on NVIDIA Jetson edge devices.
- Designed data flow specifications between Jetson devices, MQTT brokers (EMQX), NestJS backend services, local PostgreSQL databases, MinIO storage, and cloud platforms — covering both online and offline operation modes.
- Specified heartbeat telemetry requirements for CPU, GPU, RAM, VRAM, temperature, and storage metrics, and defined presence-detection thresholds to prevent false offline alerts during heavy event processing.
- Identified 10+ fault scenarios including hung MQTT connections, broker payload overflow (5 MB limit), 755 events stuck during WAN outages, and delayed media synchronization — each documented with expected recovery behavior.
- Converted clarified requirements into MQTT message structures, API behavior, database schemas, security rules (HMAC-SHA256, mTLS), OTA firmware update workflows, and testing scenarios.
- Validated that critical AI alerts were preserved and synchronized after WAN outages, confirming zero-loss behavior against the defined acceptance criteria for offline operation.

Technologies involved: NestJS, PostgreSQL, Prisma, MQTT (EMQX), mTLS, Docker, NVIDIA Jetson, MinIO, AWS

Mochi E-Learning | Gamified Vocabulary Platform Personal Project

Product Analysis, Workflow Design, and Solution Validation

- Analyzed learning objectives and defined a product scope targeting 1,000+ word retention in 90 days — designing workflows for account registration, multi-language support (Japanese & English), subscription management, course enrollment, spaced repetition review, and AI-generated vocabulary explanations.
- Specified a 4-state subscription lifecycle (Free/Trial/Active/Expired) with business rules for trial activation limits, VNPay payment processing, secure callback integrity verification, automated expiration, and free-user vocabulary caps (50-word limit).
- Designed the course enrollment and learning progress model across a 16-model database schema — defining rules for premium content gating, lesson completion tracking, automatic progress calculation, and course completion state transitions.
- Identified 15+ edge cases across payment and enrollment flows — including duplicate VNPay callbacks, interrupted checkouts leaving orphaned pending subscriptions, trial-to-paid upgrade conflicts, and inconsistent access states after subscription expiration.
- Defined acceptance criteria for 6 gamified practice modes (matching, multiple-choice, voice, typing, reading, writing) and 3 admin panels (subscription, user management, analytics) — validating each against the specified business rules end-to-end.

Technologies involved: ReactJS, NestJS, PostgreSQL, Prisma, TailwindCSS, Docker, VNPay, Gemini AI

Education

Lincoln University College, Malaysia – Bachelor of Information Technology 2023 – 2027 (Expected)

Awards

Certificate of Appreciation, PathTech – Excellence in Client Delivery January 2026

Additional Information

Interests: Product Requirements Engineering, Process Optimization, IoT System Analysis, Data-driven Decision Making, AI-assisted Workflows.